Lecture 9

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A Recap

- So far, portfolio theory has been based on the Markowitz mean–variance framework
- Investors choose portfolios based on:
  - Expected return
  - Variance (risk)
  - Efficient frontier summarizes optimal tradeoffs
- With a risk-free asset, all investors hold:
    - the same optimal risky portfolio
    - combined with the risk-free asset in different proportions

The Estimation Problem

- Implementing Markowitz requires estimating:
  - $E(R_i)$ for all assets
  - σi_2 for all assets
  - Cov(R_i,R_j) for all asset pairs
- Total parameters:

$$N \text{ expected returns} + \frac{N(N+1)}{2} \text{ variances/covariances}$$

- Grows very quickly as N increases

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Practical Consequences

- Expected returns are noisy and difficult to estimate
- Small errors in inputs mean large changes in optimal weights
- Mean–variance portfolios often exhibit:
  - extreme long and short positions
  - high leverage
  - poor out-of-sample performance
- The theory is elegant, but implementation is fragile

Realized Returns vs Expectations

- Asset returns are uncertain
- Any realized return can be written as:
  - R_i=$E(R_i)$ + unexpected return
- Expected return reflects:
  - compensation for risk
  - information known at time 0
- Unexpected return reflects:
  - new information
  - surprises
- If something were predictable, it would already be in the expectation

The Unexpected Return

- Unexpected return captures news arriving during the period
- Sources of news:
  - economy-wide events
  - firm-specific events
- We can decompose the surprise:
  - R_i - $E(R_i)$ = systematic shock + firm-specific shock

Two Types of Risk

- To reiterate, there are two types of risk in this context:
  - Systematic (market) risk
    - macroeconomic news
    - affects many assets simultaneously
    - cannot be diversified away
  - Firm-specific (idiosyncratic) risk:
    - earnings surprises
    - lawsuits
    - management changes
    - diversified away in large portfolios

Modeling the Surprise

- Let:
  - m = unexpected market return
  - e_i = firm-specific shock
  - Asset i's sensitivity to market news: measured by β_i
- Then:
  - $R_i - E(R_i) = \beta_i m + e_i$

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The Single-Factor Return Model

- Rearrange to combine expected return and the surprise components:
  - $R_i = E(R_i)+\beta_i m+e_i$
  - $\beta_i m$ is the systematic surprise component
  - e_i is the idiosyncratic surprise component
- Assumptions:
  - E(m)=0
  - $E(e_i)$=0
  - Cov(e_i,m)=0
  - Cov(e_i,e_j)=0 for $i \neq j$
  - Correlation across assets arises only through market shocks

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Why the Expected Surprise Is Zero

- Expectations are best forecasts given available information
- If:

$$E(R_i - E(R_i)) \neq 0$$

- Then expectations were systematically biased
- Rational investors update expectations until:

$$E(R_i - E(R_i)) = 0$$

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Market Surprises Average to Zero

- Let m denote the unexpected market return
- Good market news and bad market news are both possible
- Over time:
  - booms offset recessions
  - positive shocks offset negative shocks
- Therefore: $E(m)$=0
- If market surprises were positive on average, investors would revise expected market returns upward
- If market surprises were negative on average, investors would revise expected returns downward
- In equilibrium, Systematic surprises must average to zero

The Same Applies To Firm-Specific Surprises

- For a given firm good news and bad news both occur
- Across firms positive and negative shocks offset
- If: $E(e_i) \neq 0$
- Then the firm's expected return was mismeasured
- So, by construction:
  - surprises are unpredictable
  - unpredictable components average to zero

Why This Matters

- Zero-mean shocks imply:
  - expected returns are driven by risk, not noise
  - variance measures risk
- Leads naturally to:
  - variance decomposition
  - Diversification
  - pricing of systematic risk

Variance of Returns

- Start with the return decomposition:

$$R_i = E(R_i) + \beta_i m + e_i$$

- Constants drop out when computing variance [Var(X+c)=Var(X)]
- Variance measures dispersion around the mean, not the level
  - $E(R_i)$ is a constant:
    - same in every realization
    - does not vary across states
    - Drops when taking variance of R_i

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Variance of Returns (con.)

- Variance rule:
  - Var(X+Y)=Var(X)+Var(Y)+2Cov(X,Y)
- Independence of Risk Components:
  - Market shocks and firm-specific shocks are assumed uncorrelated
  - So Cov (m,e_i)=0
- So, dropping the constant and Cov(m,e_i)=0 give us:

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{e_i}^2$$

Covariance Between Two Securities In The Single-Factor Model

- For assets i and j:
  - $R_i = E(R_i)+\beta_i m+e_i$
  - $R_j = E(R_j)+\beta_j m+e_j$
- Both are exposed to the same market shock m
- Covariance depends only on random components:
  - $\text{Cov}(Ri,Rj)=\text{Cov}(\beta_i m+e_i,\ \beta_j m+e_j)$
- Using linearity of covariance:
  - $\text{Cov}(R_i,R_j)=\beta_i\beta_j\text{Var}(m)+\beta_i\text{Cov}(m,e_j)+\beta_j\text{Cov}(m,e_i)+\text{Cov}(e_i,e_j)$
- Assuming:
  - $\text{Cov}(m, e_i)=\text{Cov}(m, e_j)=0$
  - $\text{Cov}(e_i,e_j)=0$ for $i\neq j$
- All cross terms disappear and we get: $\text{Cov}(R_i,R_j)=\beta_i\beta_j\sigma_m^2$

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The Common Market Factor

- In the single-factor model, systematic risk is captured by:
  - m=unexpected market return
- This represents:
  - economy-wide news
  - shocks affecting many assets simultaneously
- The market factor:
  - drives covariance across assets
  - itself is not directly observable
- In practice, we proxy the market factor using a broad market index

Using a Market Index as a Proxy

- In practice, we proxy the market factor using a broad market index
- Examples:
  - S&P 500
  - CRSP value-weighted index
  - MSCI World
- Rationale:
  - diversified portfolio
  - firm-specific shocks largely cancel out
  - index movements primarily reflect systematic news

From Single-Factor Model to Single-Index Model

- Recall the single-factor return decomposition:
  - $R_i = E(R_i) + \beta_i m + e_i$
- To make the model operational, we proxy the market factor using a broad market index.
- This leads to the single-index model:
  - $r_i = \alpha_i + \beta_i r_M + e_i$
- where:
  - r_M =excess return on the market index
  - β_i =exposure to systematic (market) risk
  - e_i =firm-specific excess return

Why We Use Excess Returns in This Model

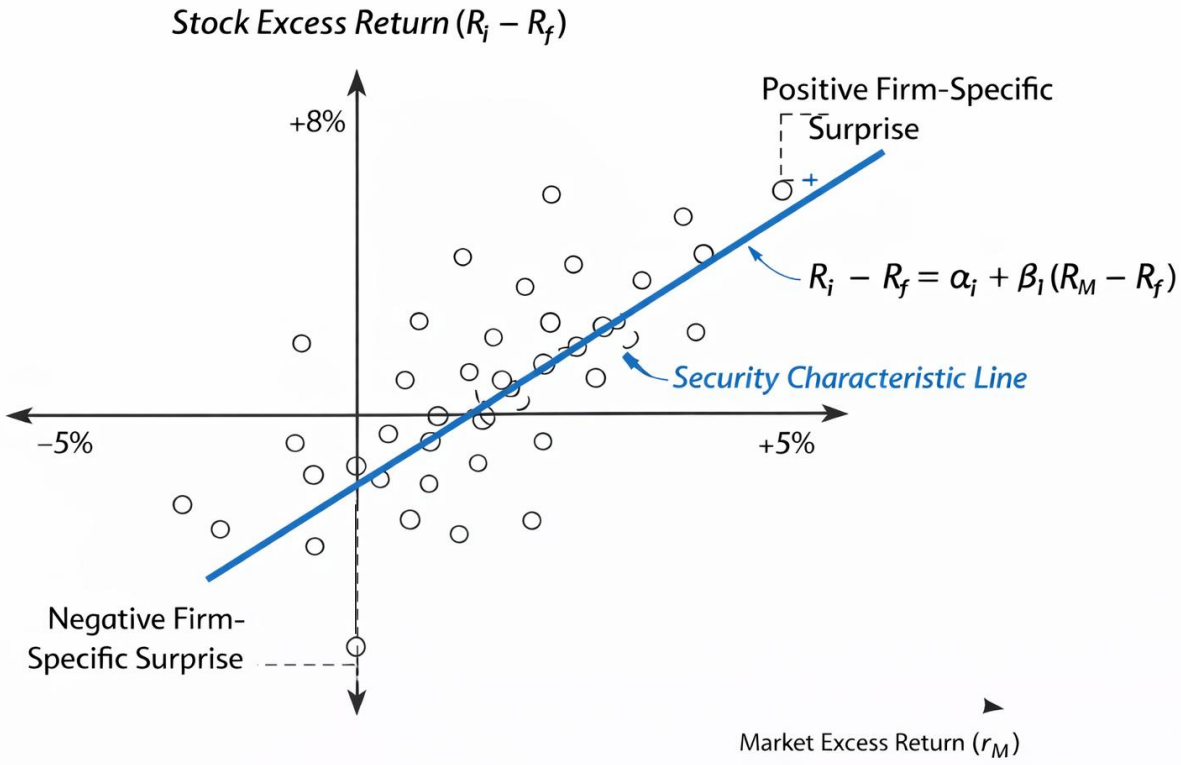
- In this model, we work with excess returns rather than total return
- Investors always have the option to invest in the risk-free asset
- A rational investor will shift from the risk-free asset to a risky asset only if the risky asset delivers a positive excess return
- Therefore, the performance of a risky investment is better measured by excess return, not total return.
- Define excess returns as:
  - $r_i = R_i - R_f$
  - $r_M = R_M - R_f$
- Excess returns isolate compensation for bearing risk from the time value of money

Empirical Implementation

- We observe:
  - stock excess returns r_i
  - market excess returns r_M
- We want to estimate:
  - β_i
  - α_i
- This leads naturally to a regression of stock excess returns on market excess returns

Visual Interpretation

- Consider a scatter plot:
  - x-axis: market excess return r_M
  - y-axis: stock excess return r_i
- Each point represents:
  - one time period
  - one realization of market and firm-specific news
- The relationship will be summarized by a fitted line



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Expected Excess Return, Beta, and Alpha

- The Security Characteristic Line (SCL) summarizes the return relationship:
  - $r_i = \alpha_i + \beta_i r_M + e_i$
- Beta (β_i)
  - slope of the SCL
  - measures sensitivity to market movements
  - captures systematic risk exposure
- Alpha (α_i)
  - intercept of the SCL
  - average excess return not explained by the market
  - measures abnormal performance
- Firm-specific shocks satisfy:
  - $E(e_i) = 0$
- Taking expectations:
  - $E(r_i) = \alpha_i + \beta_i E(r_M)$

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Set of Estimates Required: Single-Index Model

- To implement the single-index model in portfolio analysis, we estimate a small set of parameters.
- Market-level estimates
  - $E(r_M)$: expected excess return on the market index
  - ${\sigma_M}^2$: variance of the market index excess return
- Asset-level estimates (for each asset i)
  - β_i: sensitivity to the market index
  - $\sigma^2(e_i)$: firm-specific (idiosyncratic) variance
  - α_i: abnormal expected excess return
- Key model restriction:
  - All comovement across assets is generated by common exposure to the market index
  - Firm-specific risk affects variance but not covariance

Number of Estimates: Index Model vs. Markowitz

Markowitz mean–variance model

- N expected returns
- $\frac{N(N-1)}{2}$ covariances
- Total Parameters:

$$\frac{N(N+3)}{2}$$

- Estimation burden grows quadratically with N

Single-index model

- N alphas (α_i)
- N betas (β_i)
- N firm-specific variances ($\sigma^2(e_i)$)
- 1 market risk premium $E(r_M)$
- 1 market variance ($\sigma_M{}^2$)
- Total Parameters: $3N+2$
- Estimation burden grows linearly

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Regression: A Simple Example (Single-Index Model)

- Single-index model (excess returns):
  - $r_{i,t} = \alpha_i + \beta_i r_{M,t} + e_{i,t}$
- For given α_i and β_i ,the predicted excess return is:
  - $\hat{r}_{i,t} = \alpha_i + \beta_i r_{M,t}$
  - (we write $\hat{r}_{i,t}$ because this is the predicted, not realized, return)
- The firm-specific residual (surprise) is:
  - $e_{i,t} = r_{i,t} - \hat{r}_{i,t}$
- Choose α_i and β_i to minimize the sum of squared errors (SSE):
  - $SSE(\alpha_i, \beta_i) = \sum_t (r_{i,t} - \hat{r}_{i,t})^2 = \sum_t (r_{i,t} - (\alpha_i + \beta_i r_{M,t}))^2$
- $SSE(\alpha_i, \beta_i)$ depends on two parameters:
  - intercept α_i
  - slope β_i
- How do we minimize SSE?
  - differentiate SSE w.r.t. α_i and β_i
  - set both equal to zero and solve
  - or compute directly using Excel

- Excel implementation (with excess returns):
- β_i= :SLOPE(r_i_range, r_M_range)
- α_i= :INTERCEPT(r_i_range, r_M_range)
- R^2= :RSQ(r_i_range, r_M_range)

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Estimation Error

- Regression produces estimates, not true parameters:
  - $\hat{\alpha}_i,\ \hat{\beta}_i \neq \alpha_i,\ \beta_i$
- Estimates are subject to sampling error:
  - depend on the length of the sample
  - depend on volatility of returns
- Each estimate comes with a standard error:
  - measures precision of the estimate
  - smaller standard error means a more reliable estimate
- Regression output typically reports:
  - coefficient estimate
  - standard error
  - t-statistic
  - p-value

Statistical Significance

- Tests whether a coefficient is statistically different from zero
  - based on the t-statistic or p-value
    - The t-statistic answers: “How many standard errors away from zero is the estimate?”
    - The p-value answers: “How likely is it to observe a t-statistic this large if the true value were zero?”
  - Common p-value conventions:
    - p<0.10: statistically significant at the 10% level
    - p<0.05: statistically significant at the 5% level
    - p<0.01: statistically significant at the 1% level
  - Common rule of thumb:$|t|$>2 is statistically significant

Alpha and Security Analysis in the Single-Index Model

- A key advantage of the single-index model is that it structures the input list for portfolio optimization
- It separates market-wide and security-specific information
- The model imposes a hierarchy of analysis:

1. Macroeconomic analysis:
   - estimates the market risk premium $E(r_M)$
   - Estimates market risk $\sigma_M{}^2$
2. Statistical analysis - estimates each security's:
   - beta βi
   - residual variance $\sigma^2(ei)$
3. Market-driven expected return
   - absent security analysis: $E(r_i)$=$\beta_i E(r_M)$ [reflects exposure to common economic risk only]
4. Security analysis (alpha)
   - generates security-specific expected returns: αi
   - captures incremental risk premium due to private or firm-level information

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The Market Index as an Investment Asset

- In the single-index framework, the market index is treated as an investable asset
  - We may include the market index (e.g., S&P 500) directly in the portfolio alongside actively researched securities
  - When treated as an asset, the market index has: β_M=1; α_M=0; $\sigma^2(e_M)$=0
- This framework allows us to construct a portfolio from:
  - n actively researched securities
  - plus one passive market-index asset
- Active management then consists of combining an active portfolio (based on alphas) with the passive market index

The Treynor–Black Framework

- The Treynor–Black model builds on the single-index model
- Key idea is to separate passive market exposure from active security selection
- The portfolio is constructed in two parts:
  - Passive portfolio
    - the market index
    - provides broad diversification
  - Active portfolio
    - composed of securities with nonzero alphas
    - reflects security analysis
  - Security analysis is used only to identify:
    - expected abnormal returns (α_i)
    - firm-specific risk ($\sigma^2(e_i)$)
  - Optimal investing consists of forming an optimal active portfolio then combining it with the market index
  - This framework limits estimation error, ensures consistency across analysts, and clarifies where active management adds value

Treynor-Black Optimal Risky Portfolio

- With estimates of each security's $\alpha_i + \beta_i r_M + e_i$, plus the market index risk premium $E(r_M)$ and covariance ${\sigma_M}^2$, we can generate the full set of inputs needed for portfolio construction.
- The single-index model implies the covariance structure: $Cov(R_i,R_j)=\beta_i\beta_j{\sigma_M}^2$, so we build the covariance matrix
- The Treynor-Black problem is to construct the optimal risky portfolio by combining a passive market index portfolio with an active portfolio based on security alphas.
- The alpha, beta, and residual variance in an equally weighted portfolio, is just the average of those parameters.
- However, we can also calculate these using a weighted average, too.
- The objective is to choose portfolio weights that maximize the Sharpe Ratio.
- We could use Excel's solver, but, conveniently, the optimal risky portfolio can be derived analytically.

Treynor-Black: Constructing the Active Portfolio

- After completing security analysis, begin with the active portfolio (securities only).
- Initial positions (signal-to-noise)

$$w_i^0 = \frac{\alpha_i}{\sigma^2(e_i)}$$

- Securities with:
  - higher alpha lend towards a larger position
  - higher residual variance lend towards a smaller position

Treynor-Black: Constructing the Active Portfolio (con.)

- Scale to portfolio weights: $$w_i = \frac{w_i^0}{\sum_{j=1}^{n} w_j^0}$$

- Calculate active portfolio alpha: $$\alpha_A = \sum_{i=1}^{n} w_i \alpha_i$$

- Calculate active portfolio residual variance: $$\sigma^2(e_A) = \sum_{i=1}^{n} w_i^2 \sigma^2(e_i)$$

Treynor-Black: Scaling the Active Portfolio Relative to the Market

- Set initial position of portfolio: $$w_A^0 = \frac{\alpha_A/\sigma^2(e_A)}{E(r_M)/\sigma_M^2}$$
- Calculate Beta of active portfolio: $$\beta_A = \sum_{i=1}^{n} w_i \beta_i$$
- Adjust for active's market exposure: $$w_A^* = \frac{w_A^0}{1 + (1 - \beta_A) w_A^0}$$

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Treynor-Black: Final Optimal Risky Portfolio

- Final portfolio weights: $w_M^* = 1 - w_A^*$ $\quad w_i^* = w_A^* w_i$
- Calculate expected excess return: $E(r_P) = (w_M^* + w_A^* \beta_A) E(r_M) + w_A^* \alpha_A$
- Compute the variance: $\sigma_P^2 = (w_M^* + w_A^* \beta_A)^2 \sigma_M^2 + [w_A^* \sigma(e_A)]^2$
- Now you have what you need to compute the optimal allocation to the risk portfolio depending on risk aversion